

Design Option 1

Genetic map of Design Option 1 showing the arrangement of genes and regulatory elements between Loc1 and Loc3. The map includes genes like pIcaA, pQacR, RibJ53, PhiIF, pTet, RibJ54, F1, AmsR, L353P31, pPhiIF, YFP_cassette, pTet, pTac, EivJ, I1, IcaA, pAmsR, pBAD, RibJ60, Q2, and QacR. It also shows regulatory elements like pIcaA, pQacR, pTet, pTac, pAmsR, and pBAD. The map is color-coded to show different genetic elements and their regulatory interactions.

Design Option 2

Genetic map of Design Option 2. The map shows three loci: Loc1, Loc2, and Loc3. Loci are represented by horizontal lines. Genes and regulatory elements are shown as colored arrows and ovals. Arrows indicate the direction of transcription. Ovals represent regulatory elements like promoters and ribosome binding sites. Colored lines connect regulatory elements to the genes they control.

- Loc1:** Contains a single gene, *pHf*, represented by a black arrow pointing right.
- Loc2:** Contains a cluster of genes and regulatory elements. From left to right:
 - pIcaRA* (pink arrow pointing right)
 - pOAcR* (green arrow pointing right)
 - RibJ53* (orange oval)
 - P3* (orange oval)
 - pHf* (black arrow pointing right)
 - ECK120033737* (black arrow pointing right)
 - pTet* (black arrow pointing right)
 - RibJ54* (blue oval)
 - F1* (blue oval)
 - AmeR* (blue arrow pointing right)
 - L353P31* (black arrow pointing right)
 - pTet* (black arrow pointing right)
 - pTlc* (black arrow pointing right)
 - ElvJ* (pink oval)
 - I1* (pink oval)
 - IcaRA* (pink arrow pointing right)
 - ECK120016170* (black arrow pointing right)
 - pAmeR* (blue arrow pointing right)
 - pBAD* (black arrow pointing right)
 - RibJ60* (green oval)
 - Q2* (green oval)
 - QacR* (green arrow pointing right)
- Loc3:** Contains a single gene, *YFP_cassette*, represented by a black arrow pointing right.

Regulatory connections (indicated by colored lines):

- pIcaRA* (pink) controls *IcaRA* (pink).
- pOAcR* (green) controls *QacR* (green).
- pHf* (black) controls *pHf* (black).
- pTet* (black) controls *AmeR* (blue) and *L353P31* (black).
- pTlc* (black) controls *ElvJ* (pink).
- pBAD* (black) controls *QacR* (green).
- pAmeR* (blue) controls *AmeR* (blue).
- YFP_cassette* (black) controls *pHf* (black).

Design Option 3

Genetic map of the YFP_cassette region. The map shows various genes and regulatory elements, including pIcaRA, pQacR, Rho953, P3, Phif, pIret, Rho954, F1, AmsR, L539p31, pIret, pIac, Elv, I1, IcaRA, pIcaRA, pAmsR, pBAD, Rho950, Q2, QacR, Rho951, pPhif, and YFP_cassette. Arrows indicate the direction of transcription for each gene. The map is flanked by Loc1 and Loc3, with a scale bar at the top.

Design Option 4

Genetic map of Design Option 4. The map shows the arrangement of genes and regulatory elements between Loc1 and Loc3. The genes and elements are: pIcaR, pQacR, RhoJ53, P3, pPhf, pIet, RhoJ54, F1, pIcaR, pIcaR, pPhf, YFP_cassette, pAmeR, pBAD, RhoJ60, Q2, and QacR. The map also shows regulatory elements like L359p31, pIac, Elv, I1, and L359p31. Arrows indicate the direction of transcription and regulation.

Design Option 5

Genetic map of Design Option 5 showing the arrangement of genes and regulatory elements between loci Loc1 and Loc3. The map includes genes like *picaRA*, *pOacR*, *RibJ53*, *P3*, *PhlP*, *ECK12003737*, *pPHF*, *YFP_cassette*, *pTet*, *RibJ54*, *F1*, *AneR*, *L353P31*, *pTet*, *pTac*, *BivJ*, *IL*, *IcaRA*, *ECK120016170*, *pAneR*, *pBAD*, *RibJ60*, *Q2*, *QacR*, and *ECK120010818*. Arrows indicate the direction of transcription and regulation.